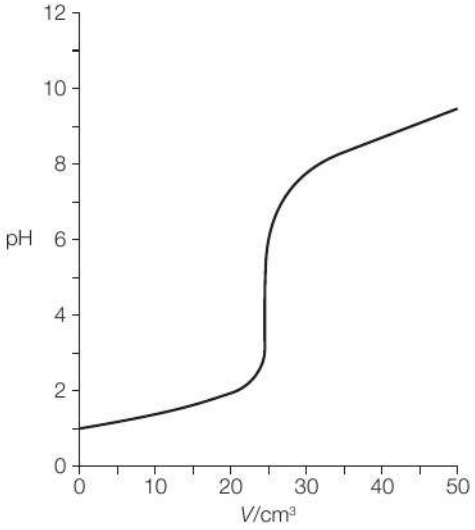
14 EXAM PRACTICE

1 In an acid–base titration, a 0.10 mol dm^{-3} solution of a base is added to 25 cm^3 of a 0.10 mol dm^{-3} solution of an acid.

The diagram shows the pH value of the solution plotted against the volume, *V*, of base added.



This diagram could represent a titration between:

A $CH_3COOH(aq)$ and $NH_3(aq)$

B $CH_3COOH(aq)$ and $KOH(aq)$

C $HCl(aq)$ and $NH_3(aq)$

D $HCl(aq)$ and $KOH(aq)$ [1]

(Total for Question 1 = 1 mark)

2 Hardness in water can be determined by titrating a sample against a reagent that forms complex ions with dissolved metal ions.

The indicator for this titration requires the pH to be maintained at about 10.

Which of these, in aqueous solution, could be used to do this?

A ammonia and ammonium chloride

B ammonium chloride and hydrochloric acid

C sodium ethanoate and ethanoic acid

D sodium hydroxide and sodium ethanoate. [1]

(Total for Question 2 = 1 mark)

3 A 1.0×10^{-2} mol dm^{-3} aqueous solution of the weak, monoprotic acid HA has a pH of 4.0.

What is the approximate pK_a value for the acid?

A 4.0 **C** 6.0 **C** 7.0 **D** 8.0 [1]

(Total for Question 3 = 1 mark)

4 The following equilibrium exists in a mixture of concentrated nitric acid and concentrated sulfuric acid:

$$HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + 2HSO_4^- + H_3O^+$$

Which statement is correct?

A HNO_3 and NO_2^+ are a conjugate acid–base pair.

B The nitric acid acts as an oxidising agent.

C The sulfuric acid acts as a dehydrating agent.

D The sulfuric acid acts as a base. [1]

(Total for Question 4 = 1 mark)

5 A sales leaflet claims that 'applications of a solution of ammonium sulfate, which is acidic, improve the growth of acid-loving rhododendron bushes by increasing the availability of nitrogen and also by increasing the pH of the soil'.

What is **wrong** with this statement?

A Aqueous ammonium sulfate is not acidic.

B Ammonium sulfate does not dissolve in water.

C To be a fertiliser, nitrogen is needed in its oxidised form (nitrate) and not in its reduced form (ammonium).

D The pH of the soil will be decreased, not increased. [1]

(Total for Question 5 = 1 mark)

6 A fruit juice contains a monobasic acid HA.

(a) The fruit juice has a hydrogen ion concentration of 2.50×10^{-4} mol dm^{-3}. Calculate the pH of the fruit juice. [2]

(b) A 25.0 cm^3 sample of the fruit juice reacted exactly with 26.70 cm^3 of 0.0100 mol dm^{-3} sodium hydroxide.

(i) Write a chemical equation for the reaction taking place. State symbols are not required. [1]

(ii) Calculate the concentration, in mol dm^{-3}, of HA in the fruit juice. [3]

(iii) Compare and contrast your answer in (b)(ii) with the hydrogen ion concentration. Then, make a deduction about the strength of the acid, HA, in the fruit juice. [2]

(c) (i) Write an equation to represent the dissociation of HA into its ions in aqueous solution. [1]

(ii) Write an expression for the acid dissociation constant, K_a, of HA(aq). [1]

(iii) The value of K_a for HA(aq) is 6.00×10^{-5} mol dm^{-3}. Calculate the concentration of the undissociated acid under these conditions. [2]

(Total for Question 6 = 12 marks)

7 A mixture of ethanoic acid, CH_3COOH, and its sodium salt CH_3COONa, can act as a buffer solution.

(a) State what is meant by the term 'buffer solution'. [2]

(b) Explain how a mixture of ethanoic acid and sodium ethanoate acts as a buffer. [4]

(c) A buffer solution is made by adding a solution of sodium hydroxide, of concentration 1.00 mol dm^{-3}, to a sample of 1.00 mol dm^{-3} ethanoic acid until half of the amount of acid present has reacted. Calculate the pH of this buffer solution.

[K_a of ethanoic acid at the temperature used is 1.70×10^{-5} mol dm^{-3}] [3]

(Total for Question 7 = 9 marks)

8 The values of the ionic product of water, K_w, are 1.00×10^{-14} mol^2 dm^{-6} at 298 K and 5.48×10^{-14} mol^2 dm^{-6} at 323 K.

(a) Calculate the pH of water at each of these two temperatures. [4]

(b) Using your answers to (a), comment on the validity of the following statement:
'Pure water is neutral because it has a pH of 7'. [2]

(c) Show that the data supplied can be used to deduce the sign of ΔH for the dissociation of water into ions. [2]

(Total for Question 8 = 8 marks)

9 In 1923, Johannes Brønsted and Thomas Lowry proposed independently a theory that when an acid reacts with a base the acid forms its conjugate base. The theory is known as the 'Brønsted–Lowry' theory.

(a) State what is meant by the term **conjugate base**. [1]

(b) (i) In each of the two equations, identify the species on the left-hand side of the equation that is behaving as a Brønsted–Lowry acid. [2]

Equation 1: $C_6H_5COO^- + HF \rightleftharpoons C_6H_5COOH + F^-$

Equation 2: $C_6H_5COOH + CN^- \rightleftharpoons C_6H_5COO^- + HCN$

(ii) Explain the relative strengths of the three acids involved in the two equilibria. [Assume that both equilibria lie well to the right-hand side.] [4]

(c) Liquid ammonia, like water, undergoes self-ionisation, according to the following equation:

$NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH_2^-$

For each of the two substances listed, explain whether a solution in liquid ammonia would be 'acidic', 'basic' or 'neutral'.

(i) Ammonium chloride, $NH_4^+Cl^-$

(ii) Sodium amide, $Na^+NH_2^-$ [4]

(Total for Question 9 = 11 marks)

10 Sulfur dioxide reacts with water to produce sulfurous acid, H_2SO_3, which is a weak dibasic acid.

The equation for the first dissociation into ions is:

$H_2SO_3(aq) \rightleftharpoons H^+(aq) + HSO_3^-(aq)$

$K_{a(1)}(298\text{ K}) = 1.20 \times 10^{-2}$ mol dm^{-3}

(a) Calculate the value of $pK_{a(1)}$ for H_2SO_3. [1]

(b) Use $K_{a(1)}$ to calculate the approximate pH of an aqueous solution of 0.500 mol dm^{-3} H_2SO_3 at 298 K. [3]

(c) The measured pH of 0.500 mol dm^{-3} H_2SO_3 is slightly lower than that calculated in part (b). Comment on a possible reason for this difference. [1]

(d) The constant K_w has a value of 1.00×10^{-14} mol^2 dm^{-6} at 298 K.

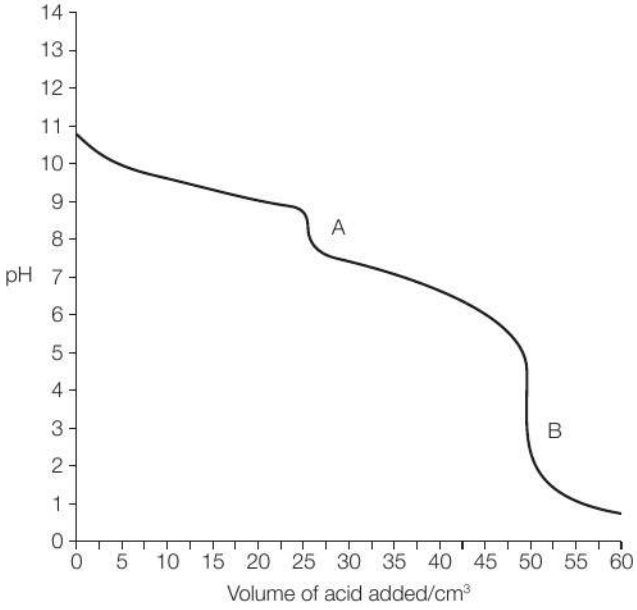
(i) Give the name of the constant K_w. [1]

(ii) Write the expression for K_w. [1]

(e) Potassium hydroxide, KOH, is a strong base in aqueous solution. Calculate the pH of 0.500 mol dm^{-3} KOH. [3]

(Total for Question 10 = 10 marks)

11 The graph shows the change in pH when dilute hydrochloric acid is titrated with sodium carbonate solution.



The overall equation for the reaction is:

$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$

(a) Suggest suitable indicators to detect the changes at A and at B. [2]

(b) Explain why the change at A would be more difficult to detect than the change at B. [2]

(c) Suggest equations for the reactions taking place at each stage. [2]

(Total for Question 11 = 6 marks)